

# Tom Silver

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## Education

### Ph.D. Massachusetts Institute of Technology

Computer Science (EECS)

Advised by Leslie Kaelbling and Joshua Tenenbaum

Thesis: *Neuro-Symbolic Learning for Bilevel Robot Planning*

Cambridge, MA  
May 2020 – May 2024

### S.M. Massachusetts Institute of Technology

Computer Science (EECS)

Thesis: *Few-Shot Bayesian Imitation Learning with Logical Program Policies*

GPA: 5.0/5.0

Cambridge, MA  
Aug 2018 – May 2020

### A.B. Harvard College

Computer Science and Mathematics

Thesis: *Luna: A Game-Based Rating System for Artificial Intelligence*

Magna cum laude with highest honors

Cambridge, MA  
Aug 2012 – May 2016

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## Employment

### Princeton University

Assistant Professor

ECE Department

Princeton, NJ  
July 2025 – Present

### Cornell University

Postdoctoral Fellow

EmPRISE Lab (PI: Tapomayukh "Tapo" Bhattacharjee)

CS Department

Ithaca, NY  
July 2024 – June 2025

### Boston Dynamics AI Institute

Research Intern

Manager: Jenny Barry

Cambridge, MA  
May 2023 – May 2024

### Google Brain Robotics

Research Intern

Managers: Shane Gu and Saminda Abeyruwan

Mountain View, CA (Remote)  
Jun – Aug 2021

### Vicarious AI

Research Engineer; Researcher

Union City, CA  
Jul 2016 – Aug 2018

### Google

Software Engineering Intern

Cambridge, MA  
Jun – Aug 2014

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## Conference Publications

1. Li, B., **Silver, T.**, Scherer, S., Gray, A. Bilevel Learning for Bilevel Planning. *Robotics: Science and Systems (RSS)* 2025.

2. Jenamani, R., **Silver, T.**, Dodson, B., Tong, S., Song, A., Yang, Y., Liu, Z., Howe, B., Whitneck, A., Bhattacharjee, T. FEAST: A Flexible Mealtime-Assistance System Tackling In-the-Wild Scenarios. *Robotics: Science and Systems (RSS)* 2025.
3. Liang, Y., Kumar, N., Tang, H., Weller, A., Tenenbaum, J., **Silver, T.**, Henriques, J., Ellis, K. VisualPredicator: Learning Abstract World Models with Neuro-Symbolic Predicates for Robot Planning. *International Conference on Learning Representations (ICLR)* 2025. **Spotlight presentation (Top 5%).**
4. Ye, R., Chen, S., Yan, Y., Yang, J., Ge, C., Barreiros, J., Tsui, K., **Silver, T.**, Bhattacharjee, T. CART-MPC: Coordinating Assistive Devices for Robot-Assisted Transferring with Multi-Agent MPC. *ACM/IEEE International Conference on Human-Robot Interaction (HRI)* 2025.
5. Liu, Z., Ju, Y., Da, Y., **Silver, T.**, Thakkar, P., Li, J., Guo, J., Dimitropoulou, K., Bhattacharjee, T. GRACE: Generalizing Robot-Assisted Caregiving with User Functionality Embeddings. *ACM/IEEE International Conference on Human-Robot Interaction (HRI)* 2025.
6. Kumar, N.<sup>\*†</sup>, **Silver, T.\***, McClinton, W.<sup>†</sup>, Zhao, L., Proulx, S., Lozano-Perez, T., Kaelbling, L., Barry, J. Practice Makes Perfect: Planning to Learn Skill Parameter Policies. *Robotics: Science and Systems (RSS)* 2024.
7. **Silver, T.**, Dan, S., Srinivas, K., Tenenbaum, J., Kaelbling, L., Katz, M. Generalized Planning in PDDL Domains with Pretrained Large Language Models. *AAAI Conference on Artificial Intelligence (AAAI)* 2024.
8. Kumar, N.<sup>\*†</sup>, McClinton, W.<sup>\*†</sup>, Chitnis, R., **Silver, T.**, Lozano-Perez, T., Kaelbling, L. Learning Efficient Abstract Planning Models that Choose What to Predict. *Conference on Robot Learning (CoRL)* 2023.
9. Li, A.<sup>†</sup>, **Silver, T.** Embodied Active Learning of Relational State Abstractions for Bilevel Planning. *Conference on Lifelong Learning Agents (CoLLAs)* 2023. **Oral presentation (Top 12).**
10. **Silver, T.\***, Chitnis, R.<sup>\*</sup>, Kumar, N.<sup>†</sup>, McClinton, W.<sup>†</sup>, Lozano-Perez, T., Kaelbling, L., Tenenbaum, J. Predicate Invention for Bilevel Planning. *AAAI Conference on Artificial Intelligence (AAAI)* 2023. **Oral presentation.**
11. **Silver, T.**, Athalye, A.<sup>†</sup>, Tenenbaum, J., Lozano-Perez, T., Kaelbling, L. Learning Neuro-Symbolic Skills for Bilevel Planning. *Conference on Robot Learning (CoRL)* 2022.
12. Chitnis, R.<sup>\*</sup>, **Silver, T.\***, Tenenbaum, J., Kaelbling, L., Lozano-Perez, T. Learning Neuro-Symbolic Relational Transition Models for Bilevel Planning. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* 2022.
13. Yang, R.<sup>\*†</sup>, **Silver, T.\***, Curtis, A., Lozano-Perez, T., Kaelbling, L. PG3: Policy-Guided Planning for Generalized Policy Generation. *International Joint Conference on Artificial Intelligence (IJCAI)* 2022.
14. Gehring, C.<sup>\*</sup>, Asai, M.<sup>\*</sup>, Chitnis, R., **Silver, T.**, Kaelbling, L., Sohrabi, S., Katz, M. Reinforcement Learning for Classical Planning: Viewing Heuristics as Dense Reward Generators. *International Conference on Automated Planning and Scheduling (ICAPS)* 2022.

15. Curtis, A., **Silver, T.**, Tenenbaum, J., Lozano-Perez, T., Kaelbling, L. Discovering State and Action Abstractions for Generalized Task and Motion Planning. *AAAI Conference on Artificial Intelligence (AAAI)* 2022.
16. **Silver, T.\***, Chitnis, R.\*, Tenenbaum, J., Kaelbling, L., Lozano-Perez, T. Learning Symbolic Operators for Task and Motion Planning. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* 2021. **Best paper finalist (Top 5).**
17. **Silver, T.\***, Chitnis, R.\*, Curtis, A., Tenenbaum, J., Lozano-Perez, T., Kaelbling, L. Planning with Learned Object Importance in Large Problem Instances Using Graph Neural Networks. *AAAI Conference on Artificial Intelligence (AAAI)* 2021.
18. Chitnis, R.\*, **Silver, T.\***, Tenenbaum, J., Lozano-Perez, T., Kaelbling, L. GLIB: Efficient Exploration for Relational Model-based Reinforcement Learning via Goal-Literal Babbling. *AAAI Conference on Artificial Intelligence (AAAI)* 2021.
19. Chitnis, R.\*, **Silver, T.\***, Kim, B., Tenenbaum, J., Lozano-Perez, T., Kaelbling, L. CAMPs: Learning Context-Specific Abstractions for Efficient Planning in Factored MDPs. *Conference on Robot Learning (CoRL)* 2020. **Plenary talk (Top 20).**
20. Zhi-Xuan, T., Mann J. L., **Silver, T.**, Tenenbaum, J., Mansinghka, V. K. Online Bayesian Goal Inference for Boundedly-Rational Planning Agents. *Conference on Neural Information Processing Systems (NeurIPS)* 2020.
21. **Silver, T.**, Allen, K., Lew, A., Kaelbling, L., Tenenbaum, J. Few-Shot Bayesian Imitation Learning with Logical Program Policies. *AAAI Conference on Artificial Intelligence (AAAI)* 2020. Earlier versions at RLDM 2019 and ICLR SPiRL Workshop 2019.
22. Loula, J., Allen, K., **Silver, T.**, Tenenbaum, J. Learning Constraint-Based Planning Models from Demonstrations. *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)* 2020.
23. Xia, V.\*, Wang, Z.\*, Allen, K., **Silver, T.**, Kaelbling, L. Learning Sparse Relational Transition Models. *International Conference on Learning Representations (ICLR)* 2019.
24. Loula, J., **Silver, T.**, Allen, K., Tenenbaum, J. Discovering a Symbolic Planning Language from Continuous Experience. *Annual Meeting of the Cognitive Science Society (CogSci)* 2019.
25. Stark, M., Schlegel, A., Wendelken, C., Park, D., Purdy, E., **Silver, T.**, Phoenix, S., George, D. Behavior is Everything – Towards Representing Concepts with Sensorimotor Contingencies. *AAAI Conference on Artificial Intelligence (AAAI)* 2018.
26. Kansky, K., **Silver, T.**, Mely, D. A., Eldawy, M., Lazaro-Gredilla, M., Lou, X., Dorfman N., Sidor S., Phoenix S., George, D. Schema Networks: Zero-Shot Transfer with a Generative Causal Model of Intuitive Physics. *International Conference on Machine Learning (ICML)* 2017.

## Journal Publications

27. Garrett, C.R., Chitnis, R., Holladay, R., Kim, B., **Silver, T.**, Kaelbling, L., Lozano-Perez, T. Integrated Task and Motion Planning. *Annual Review of Control, Robotics, and Autonomous Systems*. Vol. 4 2021.

28. Colubri, A.\*, **Silver, T.\***, Fradet, T., Retzepi, K., Fry, B., Sabeti, P. Transforming Clinical Data into Actionable Prognosis Models: Machine-Learning Framework and Field-Deployable App to Predict Outcome of Ebola Patients. *PLoS Neglected Tropical Diseases* 2016.

## Workshop Publications

29. **Silver, T.\***, Hariprasad, V.\*<sup>†</sup>, Shuttleworth, R.\*<sup>†</sup>, Kumar, N.<sup>†</sup>, Lozano-Perez, T., Kaelbling, L. PDDL Planning with Pretrained Large Language Models. *Workshop on Foundation Models for Decision Making (FMDM) @ NeurIPS 2022*.
30. **Silver, T.\***, Chitnis, R.\*, Kumar, N.<sup>†</sup>, McClinton, W.<sup>†</sup>, Lozano-Perez, T., Kaelbling, L., Tenenbaum, J. Inventing Relational State and Action Abstractions for Effective and Efficient Bilevel Planning. *Multi-disciplinary Conference on Reinforcement Learning and Decision Making (RLDM) 2022*. Short version of “Predicate Invention for Bilevel Planning” (AAAI 2023). **Spotlight talk.**
31. Zeng, C.<sup>†</sup>, **Silver, T.** Learning Search Guidance from Failures with Eliminator Edge Sets. *Workshop on Bridging the Gap Between AI Planning and Reinforcement Learning (PRL) @ ICAPS 2021*.
32. **Silver, T.**, Chitnis, R. PDDL Gym: Gym Environments from PDDL Problems. *Workshop on Bridging the Gap Between AI Planning and Reinforcement Learning (PRL) @ ICAPS 2020*.
33. **Silver, T.\***, Chitnis, R.\*, Tenenbaum, J., Lozano-Perez, T., Kaelbling, L. Learning Skill Hierarchies from Predicate Descriptions and Self-Supervision. *Workshop on Generalization in Planning (GenPlan) @ AAAI 2020*.
34. Loula, J., **Silver, T.**, Allen, K., Tenenbaum, J. Learning Models for Mode-based Planning. *Workshop on Generative Modeling and Model-Based Reasoning for Robotics and AI (MBRL) @ ICML 2019*.

## Preprints and Others

35. **Silver, T.\***, Allen, K.\*, Tenenbaum, J., Kaelbling, L. Residual Policy Learning. *arXiv* 2018.
36. **Silver, T.** Luna: A Game-Based Rating System for Artificial Intelligence. *Undergraduate thesis with Professor Stuart M. Shieber, Harvard College* 2016.

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## Teaching Experience

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|-----------------------------------------------------------------------|----------------------------------|
| <b>6.s058 / 16.420: Representation, Inference and Reasoning in AI</b> | MIT                              |
| Co-instructor                                                         | Aug – Dec 2021                   |
| <b>6.882: Structured Models for AI</b>                                | MIT                              |
| Teaching Assistant                                                    | Aug – Dec 2020                   |
| <b>CS 121: Theory of Computation</b>                                  | Harvard College                  |
| Head Teaching Fellow; Teaching Fellow                                 | Sep – Dec 2014; Sep – Dec 2015   |
| <b>CS 20: Discrete Math</b>                                           | Harvard College                  |
| Head Teaching Fellow; Teaching Fellow                                 | Jan – May 2015; Jan – May 2016   |
| <b>Digital Literacy Project</b>                                       | Harvard College / Boston Schools |
| President; Teaching Volunteer                                         | Sep 2013 – May 2016              |

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## Invited Talks

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|--------------------------------------------------------------|----------|
| Summer School on Neurosymbolic Programming                   | Jun 2024 |
| Cornell Tech (Rush Group)                                    | Nov 2023 |
| Cornell University (Bhattacharjee Group)                     | Oct 2023 |
| Yale University (Scassellati Group)                          | Sep 2023 |
| Princeton University (Griffiths Group)                       | Mar 2023 |
| Rutgers University (CS Department Colloquium)                | Mar 2023 |
| Columbia University (Song Group)                             | Feb 2023 |
| George Mason University (Controls and Robotics Seminar)      | Feb 2023 |
| AAAI AI & Robotics Bridge Session                            | Feb 2023 |
| Oxford University (CS Department Seminar)                    | Jan 2023 |
| DeepMind London (Shanahan Group)                             | Jan 2023 |
| New York University (Pinto Group)                            | Jan 2023 |
| Massachusetts Institute of Technology (Andreas Group)        | Nov 2022 |
| Northeast Robotics Colloquium (NERC) (Poster)                | Oct 2022 |
| University of New Hampshire (Ruml Group)                     | Sep 2022 |
| Meta Reality Labs Research (Desai Group)                     | Sep 2022 |
| Stanford University (BEHAVIOR Group)                         | Sep 2022 |
| Technical University of Berlin (TU Berlin) (Toussaint Group) | Dec 2021 |
| CogSci 2021 Workshop: Minds At Play                          | Jul 2021 |
| Allen Institute for AI (AI2)                                 | Mar 2021 |
| Delft University of Technology (TU Delft) (Oliehoek Group)   | Feb 2021 |
| Brown University (Konidaris, Tellex, Littman Groups)         | Jan 2021 |
| Arizona State University (Srivastava Group)                  | Jan 2021 |
| Brown University (Konidaris, Tellex, Littman Groups)         | Mar 2020 |

## Fellowships, Honors, and Awards

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|------------------------------------------------------------|------------------------------|
| Outstanding MIT UROP Mentor Awards (Nominated)             | Apr 2023                     |
| IJCAI-ECAI 2022 Distinguished PC (Top 3% Reviewer)         | Sep 2022                     |
| MIT EECS Hazen Teaching Award                              | June 2022                    |
| NSF Graduate Research Fellowship                           | Aug 2018 – May 2023          |
| MIT Stata Family Presidential Fellowship                   | Aug 2018 – May 2019          |
| Blumberg Creative Science Prize (Mather House, Harvard)    | May 2016                     |
| Derek Bok Certificate of Distinction in Teaching (Harvard) | Jan 2015, May 2015, Jan 2016 |
| Harvard College PRISE Fellowship                           | May – June 2013              |
| Research Science Institute (RSI)                           | May – June 2011              |

## Workshop and Competition Organizing Committees

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|---------------------------------------------------------------------------------------------------------------|--------------|
| <b>PhyRC: Physical Robotic Caregiving Challenge @ ICRA 2025</b>                                               | Atlanta, GA  |
| Co-organizers: Ruolin Ye, Shuaixing Chen, Justin Guo, Martin Leroux, Binit Shah, and Tapomayukh Bhattacharjee | May 2025     |
| <b>Learning Effective Abstractions for Planning (LEAP) @ CoRL 2023</b>                                        | Atlanta, GA  |
| Co-organizers: Naman Shah, Eric Rosen, David Paulius, Beomjoon Kim, Georgia Chalvatzaki                       | Nov 2023     |
| <b>Learning for Task and Motion Planning @ RSS 2023</b>                                                       | Daegu, Korea |
| Co-organizers: Danfei Xu, Danny Driess, Jeannette Bohg, Rohan Chitnis, Shuo Cheng, Zhutian Yang               | Jul 2023     |
| <b>Learning, Perception, and Abstraction for Long-Horizon Planning @ CoRL 2022</b>                            | Auckland, NZ |

Co-organizers: Gregory Stein, Jana Kosecka, Rohan Chitnis, Yezhou Yang

Dec 2022

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## Advisees

|                                                           |                           |
|-----------------------------------------------------------|---------------------------|
| Nishanth Kumar (MIT PhD Student)                          | Fall 2021 – May 2024      |
| Willie McClinton (MIT PhD Student)                        | Fall 2021 – May 2024      |
| Ryan Yang (MIT Undergrad → Applied Intuition)             | Spring 2021 – May 2024    |
| Reece Shuttleworth (MIT Undergrad)                        | Summer 2022 – Spring 2023 |
| Ashay Athalye (MIT Master's Student → Boston Dynamics AI) | Fall 2021 – Spring 2023   |
| Lilian Luong (MIT Master's Student → Applied Intuition)   | Fall 2022 – Spring 2023   |
| Jagdeep Bhatia (MIT Undergrad → Berkeley PhD Student)     | Fall 2021 – Spring 2023   |
| Amber Li (MIT Master's Student → CMU PhD Student)         | Fall 2021 – Fall 2022     |
| Varun Hariprasad (RSI → MIT Undergrad)                    | Summer 2022               |
| Abraham Mitchell (University of Arkansas Undergrad, MSRP) | Summer – Fall 2022        |
| Wester J Aldarondo-Torres (UPR Undergrad, MSRP)           | Summer – Fall 2022        |
| Catherine Zeng (Harvard Undergrad → CMU PhD Student)      | Fall 2020 – Spring 2022   |
| Shariqah N Hossain (MIT Undergrad → InterSystems)         | Fall 2020 – Spring 2021   |

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## Reviewing

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|------------------------------------------------------------------------|------------------------|
| AAAI Conference on Artificial Intelligence (AAAI)                      | 2021, 2022, 2023, 2024 |
| Autonomous Robots                                                      | 2024                   |
| Conference on Robot Learning (CoRL)                                    | 2022, 2023, 2024, 2025 |
| Frontiers in Robotics and AI                                           | 2024                   |
| Workshop on Generalization in Planning (GenPlan)                       | 2021, 2022, 2023       |
| International Conference on Human-Robot Interaction (HRI)              | 2025                   |
| International Conference on Learning Representations (ICLR)            | 2021, 2022             |
| International Conference on Machine Learning (ICML)                    | 2022, 2023             |
| International Conference on Robotics and Automation (ICRA)             | 2021, 2022, 2023, 2025 |
| International Joint Conference on Artificial Intelligence (IJCAI)      | 2022, 2023             |
| International Conference on Intelligent Robots and Systems (IROS)      | 2023, 2024             |
| International Symposium on Robotics Research (ISRR)                    | 2022                   |
| Journal of Artificial Intelligence Research (JAIR)                     | 2022                   |
| Conference on Neural Information Processing Systems (NeurIPS)          | 2021, 2022, 2023       |
| Workshop on Planning and Robotics (PlanRob @ ICAPS)                    | 2023                   |
| Workshop on Bridging Planning and Reinforcement Learning (PRL @ IJCAI) | 2023                   |
| Robotics and Automation Letters (RA-L)                                 | 2021, 2022, 2023, 2025 |
| Reinforcement Learning Conference (RLC)                                | 2025                   |
| Robotics: Science and Systems (RSS)                                    | 2024, 2025             |
| Transactions on Human-Robot Interaction (THRI)                         | 2025                   |
| Transactions on Machine Learning Research (TMLR)                       | 2022, 2023             |

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